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Factors contributing to hit-and-run in fatal crashes

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ABSTRACT

Hit-and-run in a collision is a punishable offence as it delays crash notification thereby delaying emergency response which increases the likelihood of traffic fatality. To obtain a better understanding of hit-and-run behavior, a logistic regression model has been applied in this study to identify the factors that might affect the occurrence of hit-and-run in fatal crashes in California, USA. Our results show that roadway functional class, routes, traffic flow, types of roadway section, speed limit, traffic control device, functioning of traffic control device, lighting condition, roadway alignment and roadway profile are important determinants that engineers can target to reduce hit-and-run in fatal crashes. In addition, targeted traffic enforcement should be performed on weekends and nighttime.

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1. Introduction

The problem of deaths and injury as a result of road crashes is now acknowledged to be a global phenomenon with authorities in many countries concerned about the growth in the number of people killed and seriously injured on their roads. Around the world, about 1.2 million people are killed every year in traffic collisions (Peden et al., 2004). In the United States alone, almost 42,000 people were killed in traffic crashes in 2002, resulting in an annual social cost of over $230 billion (Blincoe et al., 2002). The road toll has since risen to more than 43,000 in 2005.[1]

Although the factors contributing to the frequency and severity of crashes are diverse and complex, one of the significant determinants of the fatality risk of a crash is hit-and-run behavior (Kim et al., 2008; Tay et al., 2008). Since about 35% of fatalities occur within 1–2 h of crash occurrence (Roess et al., 2004), leaving the victim at the crash scene without reporting it delays crash notification and thus medical assistance, which increases the fatality risks. Therefore, hit-and-run behavior is a punishable offence law in most jurisdictions.

One way to reduce hit-and-run crashes is to increase the level of traffic enforcement. For example, police in Escondido and other Southern Californian counties in the United States increased the number of checkpoints for drunk drivers and people driving without a license. Even though hit-and-run is not specifically targeted in the checkpoint program, the increased enforcement did result in a 20% drop in hit-and-run collisions in 1 year (Gaona and Soto, 2007). Nevertheless, it should be noted that traffic enforcement may not be able to permanently reduce risky and dangerous driving behaviors (Bjornskau and Elvik, 1992). Without a change in driver attitudes and beliefs, enforcement serves only as a restraint and once removed, drivers are likely to return to their usual behaviors. Nevertheless, it is important to provide traffic officers with intelligence on where and when to target their enforcement efforts.

To change drivers' attitudes and beliefs, it may be better to rely on driver education but the success rates of educational campaigns vary considerably and depend largely on the target audiences and the behaviors targeted (Tay, 2005, 2008). Although road safety educational campaigns have been successfully applied to change driver attitudes and reduce crashes involving drinking and driving (Elder et al., 2004; Tay, 1999, 2002, 2005), speeding (Tay, 2004; Lewis et al., 2007; Cameron et al., 2003), driving while fatigued (Tay and Watson, 2002), and use of child booster seats (Ebel et al., 2003), no publicly available study has yet been conducted to evaluate the effectiveness of educational campaigns targeted at reducing hit-and-run crashes. Moreover, since there is no reliable or comprehensive database on hit-and-run offenders available in most jurisdictions, it is difficult to design theoretically sound and evidence based educational campaigns (Tay, 2008).

Therefore, highway engineering must play a significant role in our effort to reduce hit-and-run incidents. A properly designed highway can influence driver behavior and more importantly, this influence is self-sustaining. For example, the concept of design speed is often used as a guide in designing a highway (AASHTO, 2001). Extant research has found that highway geometric design, traffic control and roadside landscape have a significant effect

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[1] http://www-fars.nhtsa.dot.gov/Main/index.aspx.

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on driver choice of speed (Fitzpatrick et al., 2001; Milton and Mannering, 1998; Tay and Churchill, 2007; Martens et al., 1997). However, relative little research has been conducted to identify the factors that influence drivers' decision to run after the crash (Tay et al., 2008; Kim et al., 2008).

The main objective of this study is to identify the factors that influence drivers' decisions to leave the crash scene without reporting the incident. It will focus on the engineering aspects of the roadway including roadway geometric features, traffic control, operational characteristics as well as crash and environmental aspects using data from the American State of California. In this paper, previous researches examining hit-and-run crashes are first reviewed. Next, an overview of the logistic regression model developed is presented, followed by a description of the database and model variables. The results of the calibrated logistic model are then examined and discussed. Finally, some concluding comments are presented.

2. Review of the literature

Although the literature on hit-and-run crashes is fairly extensive, most of the studies concentrated on developing methods to identify the vehicles involved to aid the apprehension of the offenders who left the crash scene without reporting it. For example, several studies concentrated in measuring the color of retrieved paint fragments from crash scenes or from the clothing of victims to identify the vehicles involved in hit-and-run crashes (Baucom, 2006; Karger et al., 2001; Taylor et al., 1989; Locke et al., 1982, 1987, 1988). Many studies in the field of medical science examined the type of injury sustained by victims in hit-and-run crashes to identify the types of vehicles involved. For example, Teresinski and Madro (2001) evaluated knee joint postmortem examinations of fatal pedestrian victims of traffic accidents to deduce the location as well as identify the type of vehicles involved. Karger et al. (2001) examined different types of fractures in the victims such as wedge shaped bone fractures and fractures in cervical and lumbar spine to deduce the nature and the sequences of hit-and-run crashes.

A few studies have examined the situations or circumstances under which hit-and-run crashes occurred and only one study provided a theoretical model of the decision to run after the crash. In a recent paper, Tay et al. (2008) developed a conceptual framework to model the decision to run after the crash using a simple economic cost-benefit approach. The main determinants of not running away from the crash scene are the expected cost associate with reporting the crash while the main determinants of running are the likelihood of apprehension and the expected benefits of getting away (not caught) and expected costs of being apprehended. The model was then applied to analyze hit-and-run crashes in Singapore and would be applied in this study to explain the results obtained as well.

In terms of empirical studies on the factors contributing to hit-and-run crashes, the most widely studied influence is the role of alcohol. Solnick and Hemenway (1994) found strong evidence of the influence of alcohol on hit-and-runs by showing that a greater proportion of hit-and-run crashes occurred at night and during the weekend; the two periods when drivers were more likely to be drinking. Psychogenic reactions and alcoholic intoxication were studied in the accused drivers in hit-and-run crashes by Barbey (1995) while Solnick and Hemenway (1995) explored how victim's characteristics, driver's characteristics and circumstances of collision affected the hit-and-run choice in fatal pedestrian crashes.

Our review of the literature found very few studies that were conducted to examine the effects of highway design and traffic controls on the incidents of hit-and-run. In one study, Tay et al. (2008) studied the effects of roadway, environmental, vehicle, crash and driver characteristics on hit-and-run crashes in Singapore. However, this study was conducted in an Asian country with a highly regimented society and a very high probability of apprehension. The factors influencing the decision to run after the crash may thus be quite different from those in a western country like the United States of America or Canada.

In another study, Kim et al. (2008) used rough set analysis tools combined with logistic regression to understand the key human factors as well as roadway features associated with hit-and-run collisions in Hawaii. Although this study is conducted using an American State, Hawaii has many unique features, such as a tropical island state, that may not be applicable to the mainland United States or other western countries. Therefore, it would be interesting to compare the results obtained in our study of hit-and-run in fatal crashes from the State of California with those obtained by Kim et al. (2008) and Tay et al. (2008) for crashes of all severity in island states.

3. Methodology

Collision data from the fatality analysis reporting system (FARS) are used in this study. Among all the states in the United States, California has a large share of the road safety problem, accounting for about 10% of total fatal crashes that occurred from 1994 to 2005. Moreover, 3420 out of 42,420 fatal crashes from 1994 to 2005 were hit-and-run crashes which made California the state with the highest proportion (8.1%) of hit-and-run crashes in the Unites States. Hence, the state of California will be used as an illustration to delineate the factors contributing to hit-and-run in fatal crashes.

In our study, the response variable, hit-and-run, is a binary or dichotomous variable. Therefore, the logistic regression is a suitable technique to use because it is developed to predict a binary dependent variable as a function of predictor variables. The logistic regression model is widely used in road safety studies where the dependent variable is binary (Hooper et al., 2006; Asbridge et al., 2005; Yan et al., 2005; Valent et al., 2002; Simoncic, 2001; Zhang et al., 2000; Shibata and Fukuda, 1994; Jones and Whitfield, 1988; Lui et al., 1988). In this model, the logit is the natural logarithm of the odds or the likelihood ratio that the dependent variable is 1 (hit-and-run crash) as opposed to 0 (reported crash). The probability P of a hit-and-run crash is given by

$$Y = \ln\left(\frac{P}{1-P}\right) = \beta X \tag{1}$$

where β is a vector of parameters to be estimated and X is a vector of independent variables. When an independent variable x_i increases by one unit, with all other factors remaining constant, the odds increase by a factor $\exp(\beta_i)$ which is called the odds ratio (OR) and ranges from zero to positive infinity. It indicates the relative amount by which the odds of the outcome (hit-and-run) increase (OR > 1) or decrease (OR < 1) when the value of the corresponding independent variable increases by one unit.

To develop the logistic model, it is necessary to pre-select various factors that are expected to influence hit-and-run behavior in fatal crashes. As mentioned previously, the focus of this paper is on examining the relationship between roadway geometric features, traffic control, operational characteristics as well as crash characteristics and environmental aspects that are expected to influence hit-and-run behavior in fatal crashes. Thus, 20 factors are selected for investigation based on the FARS database. In addition to factors that are selected based on previous studies (Tay et al., 2008; Kim et al., 2008), most of the engineering data are also included in the model due to the specific focus of this paper. Summary statistics of the variables used are reported in Table 1.

Table 1
Summary of variables.

Variables	Distribution (%)
Hit-and-run	
Yes	8.1
No	91.9
Day of week	
Weekdays	64.9
Weekends	35.1
Time of day	
Peak hours	34.2
Off peak hours	19.1
Night time	45.2
Unknown	1.5
Number of vehicles	
One	60.0
Two	32.1
Three or more	7.9
Pedestrian involvement	
Yes	24.7
No	75.3
Collision type	
Single vehicle hit object	64.8
Rear end	5.5
Head on	9.4
Angle	17.5
Side swipe	2.7
Others	0.1
Roadway functional class	
Local street	10.1
Collector road	15.1
Minor arterial	22.7
Principal arterial	21.5
Freeway	16.6
Interstate arterial	13.8
Others	0.2
Route type	
Interstate	15.0
US highway	4.5
State highway	24.3
County road	20.8
Municipal road	34.8
Frontage road	<0.1
Others	0.6
Median type	
Undivided	49.3
Divided highway: median strip and no barrier	29.7
Divided highway: median strip with barrier	16.1
One-way street	0.7
Divided highway: two-way and left-turn lane	2.5
Others	1.3
Road segment	
Mid block or link	72.4
Intersection	14.9
Intersection related	8.1
Driveway	0.3
Ramp	1.9
Rail	0.5
Others	1.9
Speed limit	
No speed limit	0.5
Speed limit ≥ 70 mph	4.9
65 ≥ speed limit > 55	17.3
55 ≥ speed limit > 45	34.6
Speed limit ≤45	41.2
Others	1.5
Types of traffic control device	
No control device	76.5
Stop	7.8
Warning sign	3.5
Signal	9.9
Others	2.3

Table 1 (*Continued*)

Variables	Distribution (%)
Status of traffic control device	
No control device	76.5
Not functioning	0.1
Functioning improperly	0.1
Functioning	23.1
Unknown	0.2
Lighting condition	
Day light	47.7
Dark	25.0
Dark but lighted	22.9
Dawn	1.9
Dusk	2.3
Unknown	0.2
Horizontal alignment	
Straight	77.6
Curved	21.6
Unknown	0.8
Vertical alignment	
Level	80.5
Grade	16.7
Hillcrest	0.3
Others	2.5
Road surface	
Dry	90.2
Wet	8.8
Snow or slush	0.3
Ice	0.2
Others	0.5
Weather	
Normal	92.6
Rain	5.4
Snow	0.2
Fog	1.3
Others	0.4
Special jurisdiction	
Yes	0.4
No	99.6
Construction zone	
Yes	2.5
No	97.5

Note that many of the contributing factors were recorded using ordered or unordered categories; thus several dichotomous variables were created to capture the effects of each of these factors, giving a total of 67 variables classified under 20 factors. For example, there were three categories used to record roadway alignment: straight, curve and unknown. Therefore, three dichotomous variables were created to capture this factor. However, one of these variables will be used as the reference case and omitted from the analyses to avoid the perfect multicollinearity problem. The estimates of the included variables will then be interpreted relative to the reference case.

Since the estimates of the included variable may be sensitive to changes in the variable(s) omitted, some researchers have chosen to retain all the variables representing the same factor if at least one of them is statistically significant (Kockelman and Kweon, 2002; Lee and Mannering, 2002; Kim et al., 2008; Tay et al., 2008) while others chose to eliminate all insignificant variables from the model to increase the efficiency of estimation (Wang and Abdel-Aty, 2008; Kim and Yamashita, 2007). For completeness, both the full model with all the important factors and variables included as well as the reduced model with only the statistically significant variables retained are reported in this study. In reducing the model, we have adopted in a stepwise reduction method using the backward reduction procedure in SPSS version 14.

Note that in contrast to the case of Singapore, where most of the offenders are caught and recorded in the database (Tay et al., 2008), most of the drivers' information and vehicles' information of the offending vehicles are missing from the FARS database as the vast majority the offending drivers who left the crash scenes without reporting them were not apprehended or their records not updated in the database. In FARS, from 1975 to 1981, if no information was known about the hit-and-run vehicle and driver, the vehicle form and driver form were not filled up. Starting in 1982, both the vehicle and driver forms were filled up and the data were identified as unknown (FARS analytic reference guide). Since the vehicles' and drivers' information of offending drivers were not available for most of cases, they have been excluded from the model in this study.

4. Results and discussions

The estimation results for the full and final logistic regression models are reported in Tables 2 and 3. In general, the models fitted the data very well; with very large chi-square statistics, very small *P*-values and relatively large pseudo-*R*-squares for a disaggregated choice models. The estimation results were fairly robust, with only a few factors experiencing significant changes (changing from statistically significant to insignificant or vice versa) in one of its categories and only the type of collision experienced a change in several of its categories. By removing head-on collision from the model, the other types of crashes (single vehicle, rear end and angle) became statistically significant. Several factors were found to be statistically insignificant in both the full and final models including roadway functional class, road surface conditions, weather conditions, special jurisdiction and construction zone. The insignificance of weather conditions was also found in similar studies in Hawaii (Kim et al., 2008) and Singapore (Tay et al., 2008) while the insignificance of road surface condition was found in Hawaii but not in Singapore. Road construction was studied in Hawaii but not in Singapore and it was also found to be insignificant in Hawaii. Finally, roadway functional class and special jurisdiction were not examined in previous studies.

Our study finds day-of-week to have a significant effect on hit-and-run behavior in fatal crashes. The negative sign of estimated coefficient ($\beta = -0.405$; $P = <0.001$) and the size of odds ratio being less than one (0.667) show that the probability of hit-and-run in fatal crashes is higher on weekends than weekdays. This result is consistent with the similar studies done by Solnick and Hemenway (1995) on hit-and-run fatal pedestrian crashes but not consistent with Tay et al. (2008), which found that day-of-week was not a significant factor in hit-and-run crashes in Singapore. Our result can be explained by the fact that the traffic volume is usually lower, and thus speed is usually higher, on weekends, resulting in more severe crashes. Moreover, the likelihood of drinking and driving, which is a serious offence, is also higher on weekends. Therefore, the cost associated with reporting the crash is expected to be higher. In addition to increasing the cost of reporting of the crash, alcohol use by drivers also increases the likelihood of running after the crash (Solnick and Hemenway, 1994; Kim et al., 2008). Finally, with lower traffic volume, drivers may also feel that they are less likely to be noticed and there is a greater chance to leave the scene quickly. It should be noted that the severity of a crash as perceived by the driver involved at the time of the crash may be different from the severity level reported in the FARS. Also, the act of running away from the crash scene without reporting it may have contributed to the subsequent fatality risk.

Time-of-day also has a significant effect on hit-and-run behavior in fatal crashes. Compared to unknown time of crash, all other time periods are found to have a decreased probability of hit-and-run in a fatal crash. This result is expected because if the

Table 2
Parameter estimates of full model.

Number of observations: 42,420
Share of hit-and-run ($y = 1$) = 8.1%
Log likelihood = −9561.5
Chi-square = 4656.5
P-value <0.0001
Pseudo-*R*-square = 0.242

Explanatory variable	Coefficient	*P*-value	Odds ratio
Constant	−4.045	<0.001	0.018
Time trend	−0.005	0.415	–
Day of week			
Weekdays	−0.405	<0.001	0.667
Time of day			
Night time	−0.833	<0.001	0.435
Off peak hours	−1.590	<0.001	0.204
Peak hours	−1.468	<0.001	0.230
Number of vehicles			
Two	0.766	<0.001	2.151
Three or more	1.290	<0.001	3.633
Pedestrian involvement			
Yes	2.303	<0.001	10.002
Collision types			
Single vehicle hit object	0.323	0.613	–
Rear end	0.853	0.182	–
Head on	−0.121	0.851	–
Angle	0.539	0.399	–
Side swipe	1.496	0.020	4.462
Roadway functional class			
Local street	0.211	0.670	–
Collector road	0.152	0.759	–
Minor arterial	0.089	0.858	–
Principal arterial	0.093	0.851	–
Freeway	0.088	0.861	–
Interstate arterial	−0.175	0.742	–
Route type			
Interstate	1.619	0.016	5.047
US highway	1.253	0.057	–
State highway	1.212	0.061	–
County road	1.529	0.018	4.613
Municipal road	1.566	0.015	4.790
Frontage road	1.120	0.390	–
Median type			
Undivided	−0.111	0.452	–
Divided highway: median strip and no barrier	−0.337	0.024	0.714
Divided highway: median strip with barrier	−0.217	0.174	–
One-way street	−0.240	0.335	–
Divided highway: two-way and left-turn lane	−0.318	0.073	–
Road segment			
Intersection	0.232	0.007	1.261
Intersection related	0.192	0.009	1.212
Driveway	−0.410	0.193	–
Ramp	0.596	<0.001	1.814
Rail	−0.733	0.237	–
Speed limit			
No speed limit	−0.649	0.123	–
Speed limit ≥ 70 mph	−0.889	<0.001	0.411
65 ≥ speed limit > 55	−0.819	<0.001	0.441
55 ≥ speed limit > 45	−0.855	<0.001	0.425
Speed limit ≤45	−0.545	<0.001	0.580
Types of traffic control device			
Stop	0.337	0.368	–
Warning sign	0.161	0.685	–
Signal	0.180	0.628	–
Others	0.019	0.953	–

Table 2 (*Continued*)

Explanatory variable	Coefficient	*P*-value	Odds ratio
Status of traffic control device			
Not functioning	−0.376	0.606	–
Functioning improperly	−2.305	0.040	0.100
Functioning	−0.432	0.229	–
Lighting condition			
Day light	0.078	0.890	–
Dark	0.431	0.440	–
Dark but lighted	0.616	0.270	–
Dawn	0.628	0.273	–
Dusk	0.377	0.508	–
Horizontal alignment			
Straight	0.241	0.343	–
Curved	−0.018	0.945	–
Vertical alignment			
Level	−0.190	0.143	–
Grade	−0.536	<0.001	0.585
Hillcrest	−0.721	0.184	–
Road surface			
Dry	0.063	0.854	–
Wet	−0.131	0.715	–
Snow or slush	−0.084	0.928	–
Ice	−0.560	0.619	–
Weather			
Normal	0.171	0.671	–
Rain	0.338	0.424	–
Snow	−0.947	0.466	–
Fog	0.228	0.602	–
Special jurisdiction			
Yes	0.506	0.286	–
Construction zone			
Yes	0.085	0.528	–

crash time is unknown, there is probably no witness available and thus a lower probability of apprehension. In addition, nighttime crashes are found to be more likely than day crashes to be associated with hit-and-run behaviors. This finding is consistent with the result obtained in previous studies on hit-and-run crashes in Singapore (Tay et al., 2008) and Hawaii (Kim et al., 2008). Also, many studies have found that nighttime has a significant influence on crash occurrence as well as severity (Kim et al., 2008; Simoncic, 2001; Valent et al., 2002; Kockelman and Kweon, 2002; Lee and Mannering, 2002). In addition to the increased likelihood of a more severe crash, and thus the higher cost of reporting the crash, nighttime crashes may also encourage running because of the perceived lower probability of being identified.

The number of vehicles involved in a fatal crash is estimated to have a significant effect on the drivers' decision to leave the scene after crash. Compared to single-vehicle crash, two-vehicle crashes (OR = 2.123; $P < 0.001$) and three or more vehicle crashes (OR = 3.572; $P < 0.001$) are associated with significantly higher probability of being a hit-and-run case in fatal crashes. This result is consistent with the finding from Singapore (Tay et al., 2008) that collisions between moving vehicles have a higher likelihood of hit-and-run behavior than single vehicle collision. Single vehicle crash drivers are less likely to leave the scene without reporting the crash because the cost of reporting the crash is expected to be lower than the cost of not reporting the crash. Even though the likelihoods of apprehension may be higher in two-vehicle and multiple-vehicle crashes due to the presence of other drivers, the likelihood that the other party surviving to bear witness may not be perceived as very high, especially in hit-and-run fatal crashes. Finally, the costs of reporting these crashes may be high enough, especially for the at fault driver, to induce running despite higher likelihood of apprehension.

Table 3
Parameter estimates of final model.

Number of observations: 42,420
Share of hit-and-run ($y = 1$) = 8.1%
Log likelihood = −9576
Chi-square = 4627.5
P-value <0.0001
Pseudo-*R*-square = 0.241

Explanatory Variable	Coefficient	*P*-value	Odds ratio
Constant	−2.882	<0.001	0.056
Day of week			
Weekdays	−0.404	<0.001	0.667
Time of day			
Night time	−0.824	<0.001	0.439
Off peak hours	−1.586	<0.001	0.205
Peak hours	−1.464	<0.001	0.231
Number of vehicles			
Two	0.753	<0.001	2.123
Three or more	1.273	<0.001	3.572
Pedestrian involvement			
Yes	2.303	<0.001	10.004
Collision type			
Single vehicle hit object	0.431	0.001	1.539
Rear end	0.966	<0.001	2.628
Angle	0.659	<0.001	1.932
Side swipe	1.616	<0.001	5.034
Route type			
Interstate	0.144	0.074	1.155
County road	0.383	<0.001	1.467
Municipal road	0.397	<0.001	1.488
Median type			
Divided highway: median strip and no barrier	−0.218	<0.001	0.804
Divided highway: two-way and left-turn lane	−0.299	0.030	0.796
Road segment			
Intersection	0.259	0.002	1.296
Intersection related	0.213	0.002	1.237
Ramp	0.631	<0.001	1.879
Speed limit			
No speed limit	−0.811	0.046	0.444
Speed limit ≥ 70 mph	−0.917	<0.001	0.400
65 ≥ speed limit > 55	−0.864	<0.001	0.421
55 ≥ speed limit > 45	−0.858	<0.001	0.424
Speed limit ≤45	−0.533	<0.001	0.575
Types of traffic control device			
Stop	0.201	0.025	1.223
Status of traffic control device			
Functioning improperly	−2.126	0.044	0.119
Functioning	−0.302	<0.001	0.740
Lighting condition			
Dark	0.349	<0.001	1.418
Dark but lighted	0.526	<0.001	1.692
Dawn	0.541	<0.001	1.719
Dusk	0.305	<0.001	1.356
Horizontal alignment			
Straight	0.230	0.001	1.258
Vertical alignment			
Grade	−0.363	<0.001	0.696

Pedestrian involvement in fatal crashes is positively related to the likelihood of hit-and-run behavior. Involvement of pedestrians in fatal crashes significantly increases (OR = 2.303; $P < 0.001$) the probability of hit-and-run in fatal crashes. This result is consistent with those obtained in a similar study in Singapore (Tay et al., 2008). The involvement of pedestrian in a crash usually results in higher perceived severity of the crash and higher expected cost of

reporting the crash. More importantly, the driver may also perceive a lower likelihood of apprehension due to lower chance of the victim being able to identify the driver or the vehicle and the lower level of noise produce when hitting a pedestrian instead of another vehicle may also attract less attention. This lack of witnesses may influence the driver to leave the crash scene without reporting it.

This study also finds that collision type affected the occurrence of hit-and-run in fatal crashes. Relative to head-on and other types of collision, single vehicle, rear end, angle and side swipe collisions are more likely to induce hit-and-run behavior in a fatal crash. Single vehicle collisions are more likely to result in hit-and-run because no other vehicle is involved in the crash. Relative to head-on collisions, the likelihood of vehicles being seriously damaged is lower in angle, side swipe and rear end crashes which increases the likelihood of leaving the crash scene and decreases the likelihood of being apprehended if the driver leaves the crash scene.

Relative to other types of roads (US highway, state highway, frontage road and other routes), interstate highway, county road and municipal road are found to increase drivers' decision to run from fatal crash scene. The relationship appears to be non-monotonic. With exception of interstate, crashes on roads managed by local governments (municipal and county) as opposed to state or federal government are associated with higher likelihood of hit-and-run in fatal crashes. These differences may reflect drivers' perceptions of the capability and persistence of enforcement at different levels of government involved. Since both Hawaii and Singapore are island states, this issue was not examined in previous hit-and-run studies.

Median characteristics are found to have a significant effect on hit-and-run behavior in fatal crashes. Relative to undivided highways and one way streets, divided highways with median strip and no traffic barrier as well as divided highways with median strip and two-way continuous left-turn lane, are found to reduce the likelihood of hit-and-run in fatal crashes. These results imply that the likelihood of hit-and-run in fatal crashes is lower on divided highways compared to undivided highways. This result may be explained by the fact that divided highways carry typically more traffic than undivided highways. Therefore, when an accident occurs on a divided highway, drivers may perceive a higher likelihood of being identified. This result is in contrast to the findings in Singapore (Tay et al., 2008) where divided road had a lower likelihood of hit-and-run compared to undivided road. However, their study includes only urban roads and also delineated expressway from the divided and undivided roads.

Our results show that the crash locations on a road segment also have a significant effect on hit-and-run in fatal crashes. Drivers involved in fatal crashes occurring at the entrance or exit of a ramp had a higher tendency to run after a crash compared to those occurring at or mid-blocks or links, a result that is consistent with the finding of Tay et al. (2008) that hit-and-run is more likely to occur at flyovers and in merging traffic. This result is expected because of the relatively higher speeds of vehicles entering and leaving a freeway which will result in more severe crashes and thus a higher cost associated with reporting the crash. Also, entry and exit ramps usually have sight distance issues which will affect both the severity of the crash and the likelihood of identification. In addition to ramps, crashes at or near intersections are also more likely to result in hit-and-run because they are more likely to be perceived as severe by the drivers involved.

The posted speed limit of a roadway is correlated with its functional class, design speed, operating speed and traffic volume. Compared to other speed regulation, all other roads have significantly lower likelihood of hit-and-run in fatal collisions. The effect of speed limit on the likelihood of hit-and-run appears to be non-linear, with crashes on roads with speed limit of 45 mph or less and no speed limit experiencing the highest likelihoods of hit-and-run. These results are likely to be related to the lower traffic volume on these roads relative to the other categories of roads, resulting in lower perceived probability of identification and apprehension.

With respect to the effect of different traffic control devices, crashes occurring at intersections with stop signs are found to be associated with lower hit-and-run behaviors. Relative to intersections with traffic signals, intersections with stop signs have lower traffic volume and therefore a lower perceived probability of identification and apprehension. Moreover, crashes occurring at locations with properly or improperly functioning traffic control devices are less likely to result in hit-and-run than crashes at locations where the traffic control devices are not functioning.

Daylight and nighttime lighting conditions are identified as significant influences on the decision to run after the crash. Relative to daylight, the likelihood of hit-and-run is higher during dawn hours (OR = 1.719; $P < 0.001$), dark but lighted condition (OR = 1.692; $P < 0.001$), fully dark condition (OR = 1.418; $P < 0.001$) and dusk (OR = 1.356; $P < 0.001$). These results are consistent with the results obtained by Tay et al. (2008) and Kim et al. (2008) which find that daylight crashes may discourage running after crashes because of the higher perceived probability of being identified.

Roadway alignment has a major effect on hit-and-run likelihood. Our study finds that the probability of hit-and-run in a fatal crash on a straight roadway is significantly higher than a crash on a curved roadway. On a curved road, drivers often reduce their speed, thereby resulting in a lower severity level when a crash occurs. Also, drivers may perceive a lower likelihood of blame on the curve and thus are more willing to report the crash. This result is consistent with the findings obtained by Tay et al. (2008) but in contrast to those obtained by Kim et al. (2008). In addition to horizontal alignment, vertical alignment was also found to have a significant effect on the likelihood of hit-and-run. The probability of hit-and-run tends to increase when a crash occurs on a road with a level profile compared to a roadway with a grade. Again, the driver may perceive a lower likelihood of blame if the crash occurs on a roadway with a grade. Since Singapore is a very flat island, no data on roadway grade was recorded. On the other hand, Kim et al. (2008) did not find any significant effect of vertical grade on the likelihood of hit-and-run in crashes in Hawaii.

5. Conclusion

Leaving the crash scene without reporting it delays crash notification and emergency medical services cannot be provided immediately. All else held constant, this delay will increase the likelihood of a fatality occurring. Therefore, it is important to understand the factors that influence a driver's decision to run after a crash so that appropriate countermeasures can be developed to prevent and reduce the incidents of hit-and-run given that a crash has occurred.

In this study, the logistic regression model was applied to identify the factors that might be associated with the occurrence of hit-and-run in fatal a crash in the American State of California. Our analysis showed that day of week, time of day, number of vehicles, pedestrian involvement, collision type, route type, median type, types of roadway segment, speed limit, traffic control device, functioning of traffic control device, lighting condition, horizontal alignment and vertical alignment were important determinants of hit-and-run behavior in a fatal crash.

In terms of traffic enforcement, more effort should be devoted to preventing risky driving behaviors, including hit-and-run, during the night and on weekends where the traffic volume is lower and the speed tends to be higher and the likelihood of detection tends to be lower. Also, enforcement efforts and CCTV installation should be targeted at intersections, especially those controlled by stop signs

where the incidents of hit-and-run are higher. In addition, roads with high speed limit (70 mph or higher) or low speed limit (below 45 mph) should also be targeted.

In terms of engineering, since divided highways are negatively correlated hit-and-run likelihood, it may be prudent to consider the erection of median separations on roadways where the incidents of these crashes are relatively high. Transportation engineers may also want to consider installing street lightings at these locations. Finally, regular inspection and maintenance of traffic control devices to ensure that they are functioning will also help in reducing the incidence of hit-and-run behaviors in crashes.

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